CLASS 11 PHYSICS: MOTION IN A PLANE

Comprehensive Formula Handbook & Exam Strategy Notes (NCERT Chapter 3)

EXAM READY
QUICK REVISION

1. VECTORS & SCALARS FUNDAMENTALS

Scalar vs. Vector Quantities

Scalar: Described completely by magnitude and unit (e.g., mass, distance, speed, temperature, energy). Follows ordinary algebra rules.

Vector: Possesses both magnitude and direction, and obeys the **triangle/parallelogram law of addition** (e.g., displacement, velocity, acceleration, force).

Unit Vector Notation

Specifies direction only with unit magnitude ($|\hat{n}| = 1$):

$$\hat{a} = \mathbf{A} / |\mathbf{A}| = \mathbf{A} / A$$

Cartesian unit vectors: $\hat{i}$, $\hat{j}$, $\hat{k}$ along x, y, and z axes.

Vector Resolution in 2D & 3D

Any vector **A** in the xy-plane can be resolved into two perpendicular rectangular components:

$$\mathbf{A} = A_x\hat{i} + A_y\hat{j} = (A\cos\theta)\hat{i} + (A\sin\theta)\hat{j}$$

Magnitude: $A = \sqrt{(A_x^2 + A_y^2)}$

Direction Angle θ: $\tan\theta = A_y / A_x \Rightarrow \theta = \tan^{-1}(A_y / A_x)$

In 3D space: $A = \sqrt{(A_x^2 + A_y^2 + A_z^2)}$

2. VECTOR ADDITION & SUBTRACTION METHODS

Analytical Method (Law of Cosines & Sines)

For two vectors **A** and **B** inclined at an angle θ:

$$\text{Resultant } R = \sqrt{(A^2 + B^2 + 2AB\cos\theta)}$$

Direction α (angle of **R** with vector **A**):

$$\tan\alpha = (B\sin\theta) / (A + B\cos\theta)$$

Sine Law: $R / \sin\theta = A / \sin\beta = B / \sin\alpha$

Component-Wise Analytical Addition

If $\mathbf{A} = A_x\hat{i} + A_y\hat{j}$ and $\mathbf{B} = B_x\hat{i} + B_y\hat{j}$, then:

$$\mathbf{R} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j}$$

Vector Subtraction: $\mathbf{R}_{sub} = \mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$

$$R_{sub} = \sqrt{(A^2 + B^2 - 2AB\cos\theta)}$$

3. KINEMATICS EQUATIONS IN TWO DIMENSIONS

Physical Quantity	Vector Form	x-Component	y-Component
Position Vector	$\mathbf{r} = x\hat{i} + y\hat{j}$	x	y
Displacement	$\Delta\mathbf{r} = \Delta x\hat{i} + \Delta y\hat{j}$	$\Delta x = x' - x$	$\Delta y = y' - y$
Instantaneous Velocity	$\mathbf{v} = d\mathbf{r}/dt = v_x\hat{i} + v_y\hat{j}$	$v_x = dx/dt$	$v_y = dy/dt$
Acceleration	$\mathbf{a} = d\mathbf{v}/dt = a_x\hat{i} + a_y\hat{j}$	$a_x = dv_x/dt = d^2x/dt^2$	$a_y = dv_y/dt = d^2y/dt^2$
Position (Const. a)	$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0t + \frac{1}{2}\mathbf{a}t^2$	$x = x_0 + v_{0x}t + \frac{1}{2}a_xt^2$	$y = y_0 + v_{0y}t + \frac{1}{2}a_yt^2$
Velocity (Const. a)	$\mathbf{v} = \mathbf{v}_0 + \mathbf{a}t$	$v_x = v_{0x} + a_xt$	$v_y = v_{0y} + a_yt$

⚡ **Key Concept: Independence of Perpendicular Motions**

Motion in a 2D plane with constant acceleration can be analyzed as **two completely independent 1D motions** along perpendicular x and y axes. Time (t) is the common link between both dimensions.

4. PROJECTILE MOTION FORMULAS & DERIVATIONS

A particle launched with initial speed v_0 at an angle θ_0 to the horizontal experiences gravity $\boldsymbol{a} = -g\hat{j}$ (assuming zero air resistance).

Parametric Equations of Motion

Initial components: $v_{0x} = v_0 \cos\theta_0$, $v_{0y} = v_0 \sin\theta_0$

Horizontal ($a_x = 0$): $x(t) = (v_0 \cos\theta_0)\, t$

Vertical ($a_y = -g$): $y(t) = (v_0 \sin\theta_0)\, t - \frac{1}{2} g t^2$

Velocity at time t: $v_x = v_0 \cos\theta_0$, $v_y = v_0 \sin\theta_0 - gt$

Trajectory Equation (Parabola)

Substitute $t = x / (v_0 \cos\theta_0)$ into y(t):

$$y = (\tan\theta_0)\, x - \left[g / (2 v_0^2 \cos^2\theta_0) \right] x^2$$

Standard form $y = ax + bx^2$ proves the path is a **parabola**.

Time of Flight (T_f)

$$T_f = (2 v_0 \sin\theta_0) / g$$

Time to reach top: $t_m = T_f / 2 = (v_0 \sin\theta_0) / g$

Maximum Height (h_m)

$$h_m = (v_0^2 \sin^2\theta_0) / 2g$$

At highest point, vertical speed $v_y = 0$, speed is $v_0 \cos\theta_0$.

Horizontal Range (R)

$$R = (v_0^2 \sin 2\theta_0) / g$$

$R_{max} = v_0^2 / g$ at $\theta_0 = 45°$

5. KEY SYMMETRY PRINCIPLES & PROJECTILE THEOREMS

Property / Phenomenon	Mathematical Relation	Exam Application / Note
Complementary Angles	R(45° + α) = R(45° − α)	Ranges are equal for angles exceeding or falling short of 45° by equal amounts α.
Horizontal Projection	Time to ground $t = \sqrt{(2h / g)}$	When projected horizontally from height h ($\theta_0 = 0°$), $v_{0y} = 0$.
Impact Speed (Horizontal)	$v = \sqrt{(v_0^2 + 2gh)}$	Speed when striking ground after throwing horizontally from height h.
Max Height vs. Range	$\tan\theta_0 = 4 h_m / R$	Direct relationship between peak height and total horizontal range.

6. UNIFORM & NON-UNIFORM CIRCULAR MOTION

Angular Kinematics Relations

For an object in a circular path of radius R:

- **Angular Speed (ω):** $\omega = \Delta\theta / \Delta t = 2\pi / T = 2\pi\nu$ (rad/s)
- **Linear Speed (v):** $v = \omega R = 2\pi R \nu$
- **Period (T) & Frequency (ν):** $T = 1 / \nu$

Centripetal Acceleration (a_c)

Always directed radially inward toward the center:

$$a_c = v^2 / R = \omega^2 R = 4\pi^2 \nu^2 R$$

In non-uniform motion: Total $a = \sqrt{(a_c^2 + a_t^2)}$ where $a_t = dv/dt$.

7. STANDARD PROBLEM-SOLVING METHODS (NUMERICAL STRATEGIES)

Case 1: Rain-Umbrella Relative Motion Problems

Problem: Rain falls vertically at $\boldsymbol{v}_r$ and wind blows horizontally at $\boldsymbol{v}_w$. Find the direction to hold an umbrella.

1. Identify vectors: Rain $\boldsymbol{v}_r = -v_r\hat{\boldsymbol{j}}$, Wind $\boldsymbol{v}_w = -v_w\hat{\boldsymbol{i}}$ (or opposite).
2. Resultant velocity: $\boldsymbol{v}_R = \boldsymbol{v}_r + \boldsymbol{v}_w$.
3. Magnitude of resultant speed: $v_R = \sqrt{(v_r^2 + v_w^2)}$.
4. Angle with vertical: $\tan\theta = v_w / v_r \Rightarrow \theta = \tan^{-1}(v_w / v_r)$. Hold umbrella at angle θ with vertical towards wind.

Case 2: Motorboat / River-Swimmer Relative Velocity

Problem: Boat velocity relative to water is $\boldsymbol{v}_{bw}$, river current velocity is $\boldsymbol{v}_w$.

1. **Net velocity of boat:** $\boldsymbol{v} = \boldsymbol{v}_{bw} + \boldsymbol{v}_w$.
2. **Law of Cosines for magnitude:** $v = \sqrt{(v_{bw}^2 + v_w^2 + 2\, v_{bw}\, v_w \cos\theta)}$ where θ is angle between boat heading and current.
3. **Direction α with heading:** $\tan\alpha = (v_w \sin\theta) / (v_{bw} + v_w \cos\theta)$.

Case 3: Height-Constrained Projectile (Ceiling / Barrier Problems)

Problem: Maximum horizontal distance without hitting a ceiling of height H.

1. Set peak height formula equal to ceiling height: $h_m = (v_0^2 \sin^2\theta_0) / (2g) \le H$.
2. Solve for maximum allowable launch angle: $\sin\theta_0 = \sqrt{(2gH)} / v_0$.
3. Calculate horizontal range using this maximum angle: $R = (v_0^2 \sin 2\theta_0) / g$.

8. QUICK REFERENCE SUMMARY OF PHYSICAL QUANTITIES

Physical Quantity	Symbol	Dimensions	SI Unit	Formula / Expression
Position Vector	**r**	[L]	m	$x\hat{\boldsymbol{i}} + y\hat{\boldsymbol{j}} + z\hat{\boldsymbol{k}}$
Average Velocity	$\bar{\mathbf{v}}$	$[L\,T^{-1}]$	$m\,s^{-1}$	$\Delta\boldsymbol{r} / \Delta t$
Instantaneous Acceleration	**a**	$[L\,T^{-2}]$	$m\,s^{-2}$	$d\boldsymbol{v} / dt = (d^2x/dt^2)\hat{\boldsymbol{i}} + (d^2y/dt^2)\hat{\boldsymbol{j}}$
Projectile Range	R	[L]	m	$(v_0^2 \sin 2\theta_0) / g$
Angular Velocity	ω	$[T^{-1}]$	$rad\,s^{-1}$	$d\theta / dt = 2\pi\nu$
Centripetal Acceleration	a_c	$[L\,T^{-2}]$	$m\,s^{-2}$	$v^2 / R = \omega^2 R$

📌 **Common Pitfalls in Board Exams**

- **Acceleration vector in UCM:** Constant in *magnitude* (v^2/R), but **NOT** constant in vector direction because it continually rotates towards the center.
- **Average Acceleration in UCM:** Over one complete cycle, average velocity change is zero, so average acceleration = **0** (null vector).